

Propagator Eval — Anchor-First Execution Plan

Shareable summary of the updated Cora-first experiment strategy

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Context. The project now uses an **anchor-dataset-first** execution strategy. The full six-dataset matrix remains in scope, but it is no longer the first operational step.

Core decision

We start with **Cora** as the anchor dataset and only scale after the analysis loop is stable.

New execution order:

- **Phase 1:** Cora intrinsic core
- **Phase 2:** Cora ablations
- **Phase 3:** Cora downstream evaluation
- **Phase 4:** Citeseer/Pubmed homophilic reproduction
- **Phase 5:** OGBN-Arxiv scalability
- **Phase 6:** Texas/Wisconsin heterophily stress test

Scientific arc:

Phenomenon → Mechanism → Reproducibility → Scalability → Failure modes

Why Cora first

- small enough for fast iteration
- standard citation benchmark
- homophilic enough for clear propagation signal
- easier debugging and visualization
- good anchor for freezing metrics, defaults, and plots

What freezes early

The first-pass intrinsic metric suite is:

- mse
- cosine_sim
- recovery_ratio
- boundary_coverage
- iteration_count
- residuals
- convergence_flag
- wall_clock_time

Policy decisions:

- boundary_coverage is **required**
- spectral_fidelity is **deferred** unless later analysis needs it
- heterophilic downstream evaluation remains in scope, but only in **Phase 6**

Phase map and configs

Phase	Purpose	Primary config(s)
1	Cora intrinsic core	phase_1_cora_intrinsic.yaml + phase_1_cora_intrinsic.yaml
2	Cora ablations / freeze defaults	phase_2_cora_ablation.yaml + runnable companion configs
3	Cora downstream evaluation	phase_3_cora_downstream.yaml
4	Homophilic reproduction	phase_4_homophilic_reproduction.yaml
5	Scalability on OGBN-Arxiv	phase_5_scalability_ogbn_arxiv.yaml
6	Heterophily stress test	phase_6_heterophily_stress.yaml

Results layout

Phase-based result roots:

```
results/  
  phase_1_cora_intrinsic/  
  phase_2_cora_ablation/  
  phase_3_cora_downstream/  
  phase_4_homophilic_reproduction/  
  phase_5_scalability_ogbn_arxiv/  
  phase_6_heterophily_stress/
```

Current runner note: the existing intrinsic/downstream runners still emit legacy per-operator/per-dataset JSON directories **inside** each phase's raw/ folder. That is acceptable for now.

Stop / go gates

- **After Phase 1:** do not scale until Cora intrinsic outputs, plots, and summary notes exist.
- **After Phase 2:** do not continue until defaults are frozen.
- **After Phase 3:** do not scale until intrinsic and downstream Cora results can be joined into one coherent narrative.
- **After Phase 4:** do not move to OGBN-Arxiv until Citeseer/Pubmed reproduction is interpretable.

Immediate operational next step

Preferred entry point:

```
cd /home/bosho/FP
```

```
python -m src.fedgnn.experiments.run_intrinsic_eval \  
  --config experiments/propagator_eval/configs/phase_1_cora_intrinsic.yaml
```

```
python -m src.fedgnn.experiments.run_intrinsic_eval \  
  --config experiments/propagator_eval/configs/phase_1_cora_intrinsic_heat_kernel.yaml
```

Write findings into:

```
experiments/propagator_eval/results/phase_1_cora_intrinsic/notes/phase_1_findings.md
```

Bottom line

The project should **not** start with the old full-matrix sweep.

The right order is:

**Cora deep understanding → Cora ablations → Cora downstream → cross-dataset reproduction → scalability
→ heterophily stress**